AI Use Impact Tracker

How the Metrics Are Computed

Sapien Labs | May 2026

https://ai-use-impact-tracker-demo.up.railway.app

This document explains every metric that appears on the AI Use Impact Tracker dashboard, including the exact formulas, denominators, suppression rules, confidence intervals, and how the Period filter pools months together.

1. Core Analytical Building Blocks

All metrics are calculated from one survey response per participant per month. Three primary inputs drive the framework.

1.1 AI-Use Frequency Scale

Each respondent indicates how often they use AI tools. Their answer is mapped to a 0–6 ordinal scale:

Level	Label	Survey response
0	Never	“I have never used an AI assistant”
1	Rarely	“Rarely”
2	Monthly	“A few times a month”
3	Weekly	“Several days a week”
4	Daily	“Several times a day”
5	Constantly	“Constantly” or equivalent
6	Always	“All of the time”

1.2 AI User Status

A respondent is classified as an AI user if their frequency level is 1 or higher (i.e., they use AI at all). Anyone at level 0 (“Never”) is a non-user.

1.3 Workplace Impact Responses

Each respondent who uses AI is asked how it has affected their work. They can select multiple answers from a list. These answers fall into three groups:

Positive impacts:

1. Improved my work quality or output
2. Created new job or income opportunities

Negative impacts:

3. Increased pressure to adapt or work faster
4. Made me worry about the future of my job or industry
5. Caused me to lose my job
6. Reduced my income or made it harder to find work

Neutral / baseline:

- No impact
- Another impact not listed
- Not sure

A respondent is tagged as having a positive impact if they selected at least one positive answer, and similarly for negative impact. These categories are not mutually exclusive — a respondent can report both positive and negative impacts.

2. Core Population Indicators

These base metrics are calculated independently for every demographic group and month.

2.1 Total Respondents (N)

The count of all individuals who completed the survey in a given demographic group and month. This serves as the denominator for the adoption rate.

2.2 Adoption Rate

The proportion of respondents who use AI at all:

$$Adoption\ \ Rate\ = \frac{Number\ \ of\ \ AI\ \ users}{N}$$

Ranges from 0 to 1 (0% to 100%).

2.3 Frequency Distribution

For each frequency level (0 through 6), we compute the share of respondents at that level. The seven shares sum to 100%.

$$Frequency\ \ Share(k)\ = \frac{Respondents\ \ at\ \ level\ \ k}{N}$$

3. The Impact Denominator

Workplace impact metrics do not use the full respondent count as their base. Instead, they use a stricter sub-population called the Impact Denominator (denoted N^d below).

To be included in the impact denominator, a respondent must meet two criteria:

7. **Active engagement:** They must be an AI user (frequency level ≥ 1).
8. **Valid response:** They must have answered the workplace impact question — including choosing "No impact" or "Not sure."

> *Why exclude non-users? Someone who has never used AI cannot report a direct workplace impact. Including them would dilute the results with zeros, making the impact shares appear artificially low.*

4. Impact Share Metrics

For each type of workplace impact, we compute the share of the impact denominator that reported it:

$$Impact\ \ Share\ = \frac{Respondents\ \ who\ \ selected\ \ that\ \ impact}{N^d}$$

The nine impact shares reported on the dashboard are:

#	Impact	Direction
1	Improved my work quality or output	Positive
2	Created new job or income opportunities	Positive
3	Increased pressure to adapt or work faster	Negative
4	Made me worry about the future of my job or industry	Negative
5	Caused me to lose my job	Negative
6	Reduced my income or made it harder to find work	Negative
7	No impact	Neutral
8	Another impact not listed	Neutral

9	Not sure	Neutral

The shares do not have to sum to 100%. Since respondents can select more than one impact, a single person can contribute to multiple shares.

5. Positive and Negative Impact Shares

Two rolled-up shares summarise the directional balance, both computed over the impact denominator:

$$Positive\ \ Share\ = \frac{Respondents\ \ reporting\ \ any\ \ \ positive\ \ impact}{N^d}$$

$$Negative\ \ Share\ = \frac{Respondents\ \ reporting\ \ any\ \ \ negative\ \ impact}{N^d}$$

A respondent counts toward the positive share if they selected impact #1 or #2 from the table above; they count toward the negative share if they selected any of #3 through #6.

6. Weighted Impact Index (WII)

The **Weighted Impact Index** is the dashboard's headline metric. Rather than treating all impacts equally, it assigns each impact a signed severity weight and then averages a per-respondent score across the impact denominator.

6.1 Severity Weights

Each workplace impact carries a weight reflecting how positive or negative that outcome is. These weights were set by the Sapien Labs team based on judgment about relative severity:

#	Workplace Impact	Weight
1	Created new job or income opportunities	**+1.00**
2	Improved my work quality or output	**+0.50**
3	Made me worry about the future of my job or industry	**−0.25**

4	Increased pressure to adapt or work faster	**−0.50**
5	Reduced my income or made it harder to find work	**−0.75**
6	Caused me to lose my job	**−1.00**
7	No impact / Another impact / Not sure	**0.00**

6.2 Per-Respondent Score

For each respondent in the impact denominator, we add up the weights of every impact they selected:

$$Personal\ \ Score\ = \sum weight\ \ of\ \ each\ \ selected\ \ impact$$

Examples:

- Selected only “Improved my work quality” → score = +0.50
- Selected “Improved quality” and “Adaptation pressure” → +0.50 + (−0.50) = 0
- Selected “Job loss” and “Reduced income” → −1.00 + (−0.75) = −1.75
- Selected “No impact” → score = 0

A single respondent’s score can go beyond ±1 if they selected several impacts in the same direction.

6.3 Computing the Index

The Weighted Impact Index is the simple average of all personal scores across the impact denominator:

$$WII\ = \frac{1}{N^d} \sum Personal\ \ Score_i$$

A positive value means the average AI user in that group had a net-positive experience; a negative value means net-negative.

6.4 Why the Weights Are Asymmetric

The total negative weight across all four negative impacts (−2.50) exceeds the total positive weight (+1.50). This is intentional: severe harms like job loss are given disproportionate weight compared to moderate benefits like improved quality.

7. Dose-Response

The “Dose-response” row on the dashboard shows the balance between positive and negative impact at each AI-use frequency level. For each level (Rarely through Always):

$$Dose\text{-}Response(k) = Positive\ Share(k) - Negative\ Share(k)$$

where the shares are computed only among respondents in the impact denominator at that specific frequency level. If a frequency level has fewer than 50 respondents, its value is shown as “n/a”.

> *Interpretation: a dose-response that climbs from +0.10 at “Rarely” to +0.40 at “Daily” suggests the net impact is more positive among heavier users — a hint of dose-dependence, not a causal claim.*

8. Confidence Intervals

Every metric is a point estimate from a sample. We compute 95% confidence intervals to quantify how precise each estimate is.

8.1 For Share Metrics (Wilson Score Interval)

All proportion-based metrics (adoption rate, each impact share, positive share, negative share) use the Wilson score interval. Given k respondents with the attribute out of n total, and z = 1.96 for 95% confidence:

$$\hat{p} = \frac{k}{n}$$

$$Adjusted\ centre = \frac{\hat{p} + z^2/2n}{1 + z^2/n}$$

$$CI = Adjusted\ centre \pm \frac{z \cdot \sqrt{[\ \hat{p}(1-\hat{p})/n + z^2/4n^2\ ]}}{1 + z^2/n}$$

The Wilson interval is preferred because it always stays within [0, 1], unlike the simpler Wald interval which can produce impossible values near the boundaries.

8.2 For the Weighted Impact Index (Wald Interval)

The WII is a continuous-valued mean, not a proportion. Each respondent’s personal score can take any value (roughly −2 to +1.5), so we use the standard mean ± margin formula:

$$Standard\ Error = \frac{\sigma}{\sqrt{N^d}}$$

$$CI = WII \pm 1.96 \times Standard\ Error$$

where σ is the standard deviation of the personal scores and N^d is the impact denominator size.

> *The 95% CI is displayed beneath the WII value on the dashboard’s hero card when viewing a single month. In rolling-window views (Last 3 / Last 6 months), confidence intervals are not shown because pooled estimates cannot preserve the original interval arithmetic.*

9. Small-Sample Suppression

Any demographic group with fewer than 50 respondents in a given month is automatically suppressed — its metrics are hidden from the dashboard. This prevents unreliable estimates from being displayed and protects the confidentiality of small groups.

The threshold applies at the most granular level. For example, if “United States, Female, 18–24” has only 41 respondents, that cell is suppressed even if “United States, Female” overall has 600.

10. Demographic Stratification

Every metric is computed independently for each combination of demographic dimensions. The dashboard supports filtering by country, gender, and age band. When you apply a filter, the dashboard loads the pre-computed values for that exact combination — it does not calculate them on the fly.

This means removing a filter doesn’t simply average the filtered cells back together; it switches to a broader pre-computed grouping that includes all respondents.

11. Rolling-Window Pooling (Period Filter)

The Period selector on the dashboard offers Single month, Last 3 months, Last 6 months, and Last 12 months. When a multi-month window is selected, the dashboard combines the monthly values using a weighted average:

$$Pooled\ \ Value_{\square} = \frac{\sum\left(monthly\ \ value\ \times\ respondent\ \ count\right)}{\sum\ respondent\ \ count}$$

For the adoption rate, the weight is the total number of respondents. For all impact-related metrics (shares, WII, dose-response), the weight is the impact denominator count.

This is an approximation. The exact pooled value would require going back to the individual-level data. The weighted-mean approach is correct in expectation but does not produce new confidence intervals, which is why CIs are hidden in multi-month views.

12. Global Summary Card

The summary box in the top-right corner of the dashboard is a fixed global reference point. It is not affected by any of the dashboard filters (month, country, gender, age). It always shows:

9. The global weighted average of the currently selected metric over the last 3 full months.
10. The total number of respondents in those 3 months.
11. The total number of respondents across the entire dataset.

A “full” month means one that has finished collecting responses. The dashboard automatically detects the most recent partial month (where data is still coming in) and excludes it from this 3-month window.

13. Known Limitations

13.1 Non-Probability Sample

The respondent pool is a convenience sample recruited via online ads. It does not use post-stratification weights. Demographic proportions in the data do not match national census proportions. All metrics describe the responding population, not the general public.

13.2 Correlation, Not Causation

Associations between AI-use frequency and impact outcomes are correlational. Higher-frequency users who report better outcomes may differ from lower-frequency users in education, occupation, or digital literacy. The dose-response display is suggestive, not causal.

13.3 Composition Effects

Month-over-month changes may reflect shifts in who is responding rather than genuine attitude change. If a recruitment campaign brings in more younger respondents one month, and younger respondents tend to be more positive about AI, the global index will rise even if no individual changed their view.

13.4 Weight Selection

The severity weights in the WII (§6) were chosen by stakeholder judgment, not derived empirically. A different set of weights would produce different index values and could alter country rankings. Sensitivity analysis has not yet been conducted.

13.5 Rolling-Window Approximation

The multi-month pooling (§11) uses a weighted average of pre-computed monthly values. This introduces minor error compared to re-computing from individual-level data, accepted as a trade-off for performance.

13.6 Data Source Filtering

The production version should restrict the respondent pool to Google Display and Meta traffic sources and down-weight organic/search traffic to 10%. This is not yet implemented; the current pipeline processes all traffic sources equally.